

VIRGINIA HIGHLANDS, VA, USA

WMO: 724058

Lat: 36.683N

Lon: 82.033W

Elev: 631

StdP: 93.97

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 53818

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.1	-7.9	-17.4	0.9	-6.8	-14.8	1.1	-4.9	9.2	5.3	8.4	4.2	2.0	260	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	32.0	22.0	30.9	21.7	29.3	21.3	24.1	28.7	23.6	27.8	23.0	26.8	2.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	19.0	25.6	22.4	18.5	25.2	22.1	18.1	24.9	76.5	28.8	73.5	27.9	71.4	26.7	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.9	5.7	DB	-14.9	33.2	3.4	1.3	-17.3	34.2	-19.3	34.9	-21.2	35.7	-23.7	36.6	(4)
(5)			WB	-15.5	25.1	3.2	0.7	-17.8	25.6	-19.7	26.0	-21.5	26.4	-23.9	27.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	2.3	4.4	8.5	13.6	18.2	22.1	23.6	23.3	20.4	14.3	8.3	4.3	(6)	
	DBStd	8.77	6.20	5.63	5.45	4.63	3.81	2.58	2.05	2.19	3.33	4.66	4.97	5.41	(7)	
	HDD10.0	827	248	169	94	21	1	0	0	0	0	16	89	189	(8)	
	HDD18.3	2294	497	389	306	151	51	5	1	1	19	139	302	436	(9)	
	CDD10.0	2159	10	13	47	128	255	363	421	413	311	148	37	12	(10)	
	CDD18.3	583	0	0	1	8	46	118	163	156	80	12	1	0	(11)	
	CDH23.3	4392	0	2	11	98	377	885	1178	1138	596	102	6	1	(12)	
	CDH26.7	1280	0	0	1	12	70	263	373	373	176	12	1	0	(13)	
(14) Wind	WSAvg	2.1	2.6	2.6	2.7	2.6	2.0	1.7	1.5	1.4	1.7	1.9	2.1	2.3	(14)	
(15) Precipitation	PrecAvg	1161	104	97	120	97	110	105	131	98	67	57	76	99	(15)	
	PrecMax	1501	171	197	202	208	201	179	206	170	134	126	157	185	(16)	
	PrecMin	810	53	37	67	41	40	42	48	34	13	5	17	38	(17)	
	PrecStd	167	31	38	42	39	40	31	35	34	35	30	35	39	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	22.3	24.8	27.9	30.1	32.4	32.9	33.0	32.3	28.0	23.7	20.8	(19)	
		MCWB	14.4	14.5	14.8	17.4	19.6	21.7	22.8	22.8	19.8	19.8	16.2	15.7	(20)	
	2%	DB	16.4	18.2	22.3	26.2	28.5	31.1	31.3	32.0	30.4	26.2	21.0	17.2	(21)	
		MCWB	12.2	12.4	14.1	16.5	19.3	21.7	22.4	22.4	20.3	18.0	14.3	12.6	(22)	
	5%	DB	13.9	15.6	19.5	24.0	27.3	29.4	30.1	30.3	28.8	23.9	18.7	14.6	(23)	
		MCWB	11.1	10.9	12.6	15.4	19.0	21.1	22.3	22.1	20.3	16.7	13.5	11.7	(24)	
	10%	DB	12.0	12.9	17.4	22.2	25.9	27.9	28.8	28.8	27.2	22.3	16.7	12.6	(25)	
		MCWB	9.9	9.2	11.5	14.7	18.5	20.7	22.0	21.5	19.8	16.5	12.1	10.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.4	16.5	16.4	19.7	22.2	24.6	25.1	25.1	23.6	22.0	17.9	16.3	(27)	
		MCDB	17.6	19.9	21.6	25.8	26.9	29.3	29.6	29.9	27.7	25.2	20.5	18.4	(28)	
	2%	WB	13.6	14.0	15.2	18.1	21.2	23.5	24.2	24.1	22.5	20.4	16.3	14.2	(29)	
		MCDB	15.6	16.9	20.3	23.4	25.9	28.0	28.7	28.8	26.5	23.7	19.4	15.9	(30)	
	5%	WB	11.8	12.0	14.0	16.9	20.3	22.7	23.6	23.4	21.6	18.9	14.7	12.6	(31)	
		MCDB	13.7	14.4	18.2	21.7	25.0	26.9	27.7	27.6	25.6	22.2	17.5	13.9	(32)	
	10%	WB	9.7	9.9	12.7	15.8	19.5	21.9	23.0	22.7	20.8	17.3	12.9	10.5	(33)	
		MCDB	11.4	12.4	16.1	20.3	24.0	26.1	26.6	26.5	24.8	20.4	15.6	12.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	10.0	11.2	12.1	11.6	11.1	10.2	10.8	11.3	11.8	11.0	9.2	(35)	
		MCDBR	12.0	13.8	14.2	14.7	13.5	13.1	11.8	12.5	13.7	14.1	13.7	11.8	(36)	
	5% WB	MCWBR	8.4	8.9	7.8	7.0	5.9	5.2	4.5	4.8	5.0	6.8	7.9	8.2	(37)	
		MCWBR	9.8	11.3	12.0	12.1	11.7	10.8	10.0	10.4	10.5	10.4	10.9	9.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.291	0.300	0.323	0.364	0.402	0.432	0.462	0.464	0.368	0.338	0.304	0.286	(40)	
		taud	2.534	2.504	2.478	2.376	2.271	2.218	2.166	2.136	2.393	2.480	2.547	2.573	(41)	
	Ebn at Noon		912	943	946	918	882	852	823	813	884	884	888	896	(42)	
		Edh at Noon	78	90	102	119	135	142	149	150	110	92	76	70	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.21	2.81	3.98	5.08	5.61	6.06	5.64	5.31	4.54	3.54	2.62	1.94	(44)	
	RadStd		0.18	0.35	0.30	0.39	0.42	0.39	0.37	0.25	0.40	0.42	0.27	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+202	+87
(47) Regional (40 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+141	+86

Nomenclature: See separate page